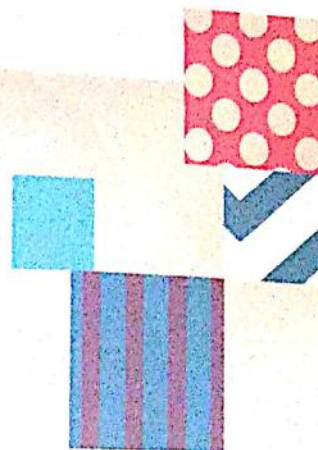
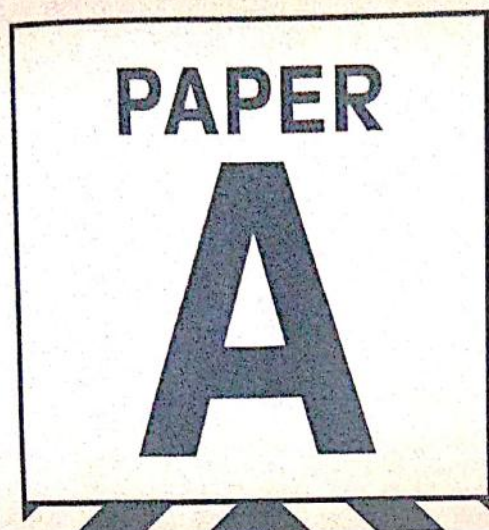




Mathematics

2019

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

35 QUESTIONS

TIME ALLOWED: 45 MINUTES

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are 35 **MULTIPLE-CHOICE QUESTIONS** (1–35).

Use the information provided to choose the **BEST** answer from the four possible options.

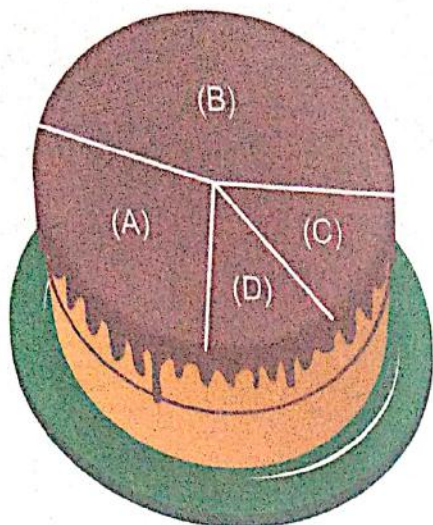
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.



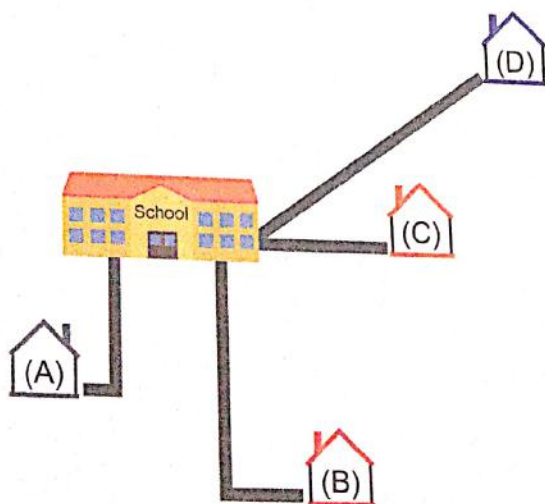
1. Amy took the largest piece of cake.
Jack took the next largest piece.



Which piece of cake did Jack take?

2. The map shows the houses of four children near a school. Each child walks to school along a path.

Which house has the shortest path to school?



3. This is the start of a number pattern.

12, 14, 16, 18, 20

What are the next two numbers in the pattern?

- (A) 21, 22
(B) 22, 24
(C) 22, 26
(D) 30, 40

4. This chart shows the shells some children collected at the beach.

Name	Shells
Peter	
Heidi	
Sue	
Henry	

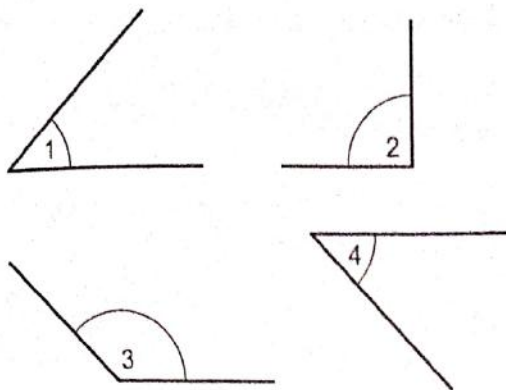
How many shells did Sue collect altogether?

- (A) 4
(B) 5
(C) 6
(D) 8

5. $54 + 23 = ?$

- (A) 59
(B) 75
(C) 77
(D) 86

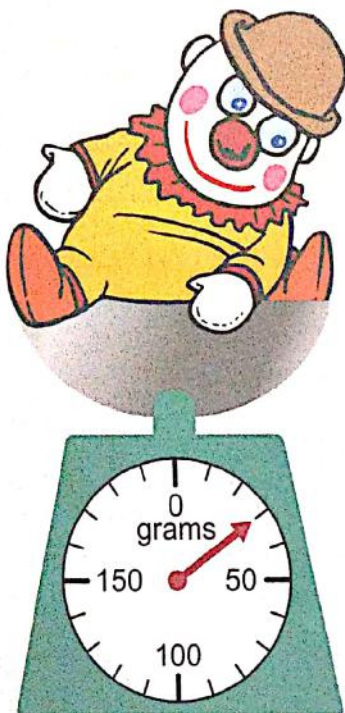
6.



Which two angles are the same size?

- (A) 2 and 3
- (B) 2 and 4
- (C) 1 and 3
- (D) 1 and 4

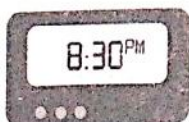
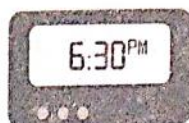
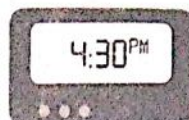
7.

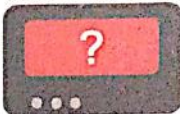


What is the mass of the clown?

- (A) 3 grams
- (B) 30 grams
- (C) 35 grams
- (D) 48 grams

8. These clocks show a pattern of times.



Which of these must replace
 to fit the pattern?

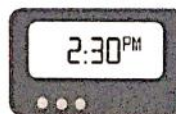
(A)



(B)



(C)



(D)



9. Toby has 21 apples. Lillian has 4 apples fewer than Toby.

How many apples does Lillian have?

- (A) 4
- (B) 17
- (C) 18
- (D) 25

10. $24 - 9$ is _____ $8 + 9$

Which of these makes the statement true?

- (A) bigger than
- (B) less than
- (C) equal to
- (D) half of

11. Lara has a robot called Speedy. Speedy sits in the middle square of a grid.

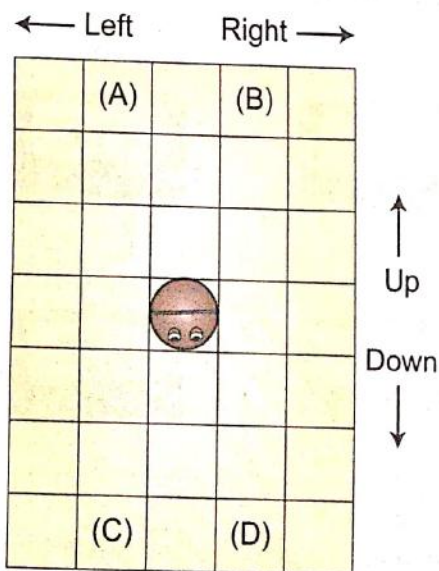
Speedy follows instructions for movement:

- U1 means move up 1 square
- D2 means move down 2 squares
- R1 means move to the right 1 square
- L2 means move to the left 2 squares

Lara gave these instructions to Speedy:

R1, D3, L2

Where will Speedy be after he has followed these instructions?



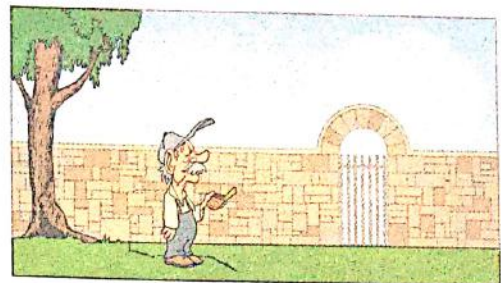
12. Liam collects rare stamps.

There are 14 stamps in half of his collection.

How many stamps does Liam have in his whole collection?

- (A) 28
- (B) 24
- (C) 12
- (D) 7

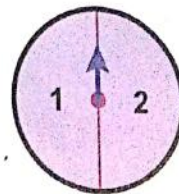
13. Mr Jones wants to measure the length of a brick wall.



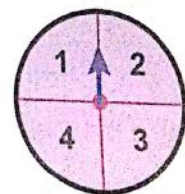
Which unit should Mr Jones use?

- (A) litres
- (B) grams
- (C) degrees
- (D) metres

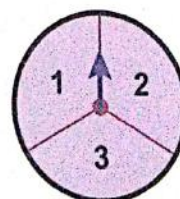
14. Which spinner has the greatest chance of landing on 2?



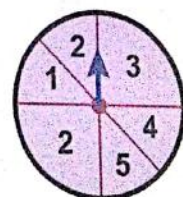
(A)



(B)



(C)



(D)

15.

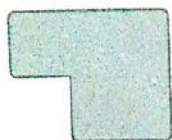


A drone bee lives for around 2 months.
This is about 2 weeks longer than a
worker bee lives.

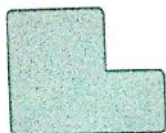
About how long does a worker bee live?

- (A) 6 weeks
- (B) 10 weeks
- (C) 15 days
- (D) 60 days

16. Anish drew this shape.



Which option shows the shape after a
quarter turn clockwise?



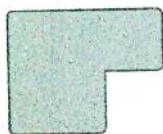
(A)



(B)



(C)



(D)

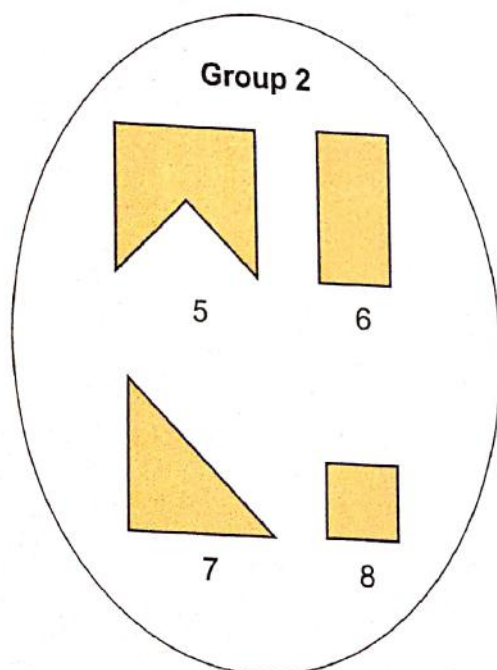
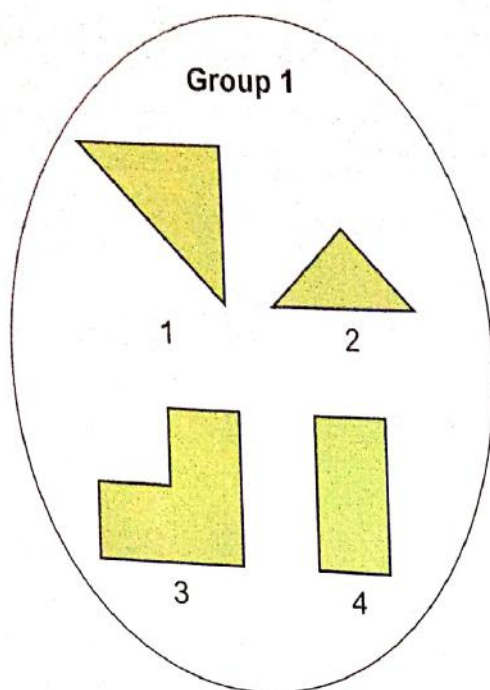
17. The table shows some information about the books some children read.

	Non-fiction	Fiction
Boys	20	16
Girls	5	15

How many children read fiction books?

- (A) 15
- (B) 16
- (C) 31
- (D) 56

18. Each shape in Group 1 fits with a shape in Group 2 to make a large square.



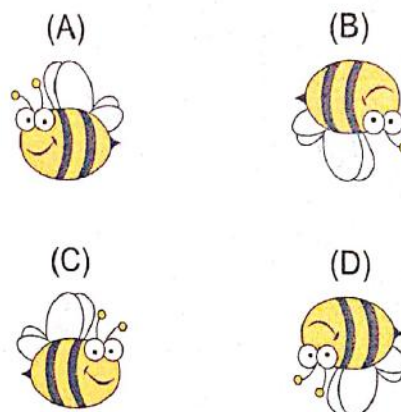
Which two shapes fit together to make a large square?

- (A) 1 and 5
- (B) 2 and 7
- (C) 3 and 6
- (D) 3 and 8

19. Sam flipped this bee across the dotted line.



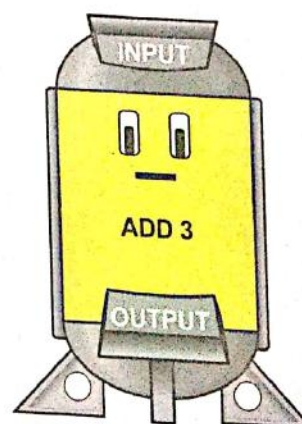
What did it look like after this?



20. Ms Green made a number machine.

This machine adds 3 to any number put into it.

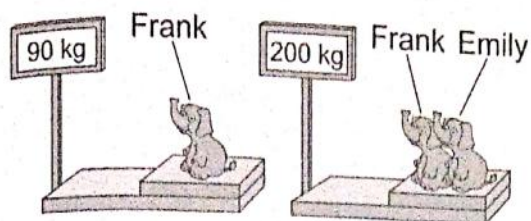
Carl put only odd numbers through the machine.



What kind of numbers would come out of the machine?

- (A) odd numbers only
- (B) even numbers only
- (C) the number 6 only
- (D) some even and some odd numbers

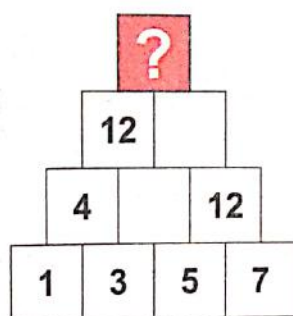
21. Emily and Frank are two baby elephants.



What is Emily's mass?

- (A) 90 kilograms
- (B) 110 kilograms
- (C) 190 kilograms
- (D) 200 kilograms

22. Sandy made a number pattern. She started on the bottom row. She added two numbers that were next to each other and wrote the answer in the box above. Some of the numbers are missing.



What must the value of ? be?

- (A) 22
- (B) 24
- (C) 32
- (D) 44

- 23.

$\text{?} + 15 = 28 - 8$

What is the missing number?

- (A) 5
- (B) 10
- (C) 20
- (D) 35

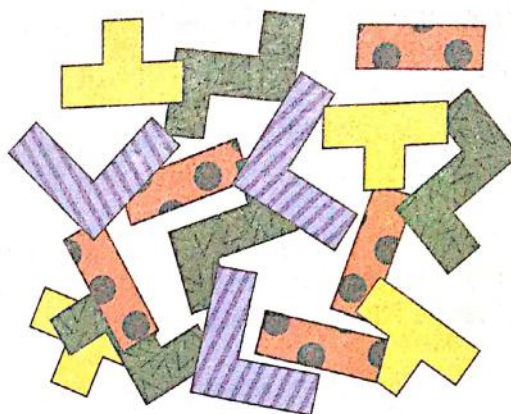
24. Emma lives in Wonderland. In Wonderland one dollar equals 8 cents.

Emma bought a toy that cost 16 Wonderland dollars.

How much did the toy cost in Wonderland cents?

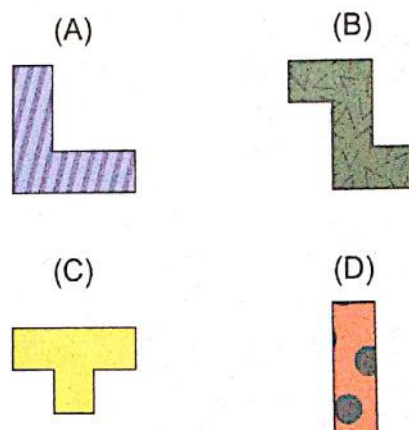
- (A) 2
- (B) 14
- (C) 24
- (D) 128

25. Andy scatters some puzzle pieces on the floor.



He picks one puzzle piece without looking.

Which piece is Andy **least** likely to pick?



26.

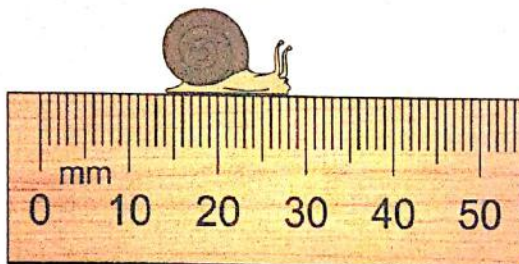
$$\text{Smiley Face} + \text{Smiley Face} = 12$$

$$\text{Smiley Face} \times \text{Smiley Face} = ?$$

What value must ? be?

- (A) 6
- (B) 36
- (C) 72
- (D) 144

27. A snail was crawling along a ruler.



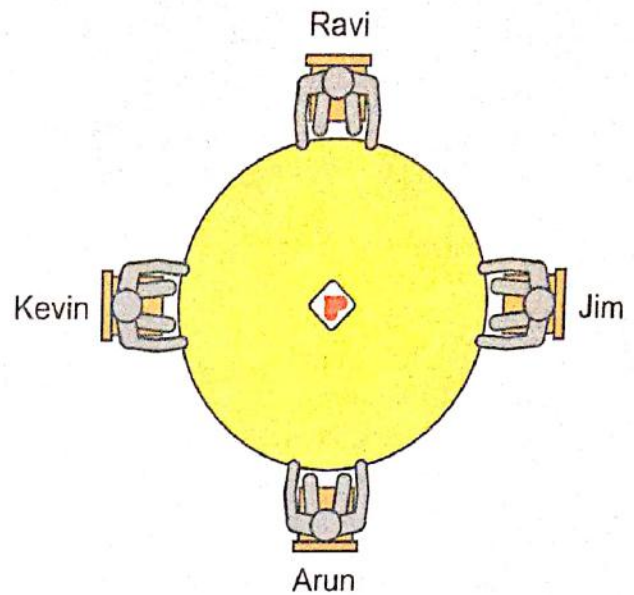
Fully stretched, the end of the snail's tail was on the 14 mm mark and its head just reached the 28 mm mark on the ruler.

Then the snail crawled to the right and stretched itself again. The end of its tail was now on the 18 mm mark.

What mark on the ruler did the snail's head reach?

- (A) 24 mm
- (B) 32 mm
- (C) 34 mm
- (D) 36 mm

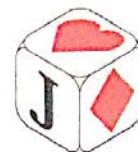
28. Four boys are sitting around a table as shown.



The picture shows how Arun, Kevin and Ravi see the cube.



Arun



Kevin



Ravi

Which of these shows how Jim sees the cube?

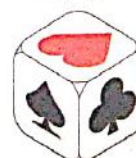
(A)



(B)



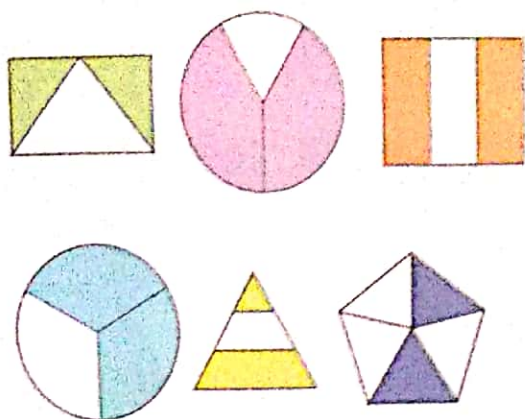
(C)



(D)



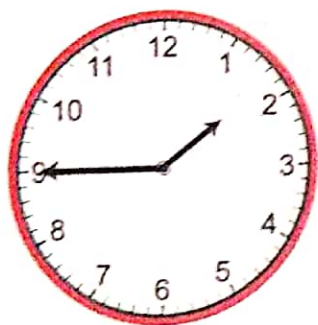
29. Priya shaded two parts in each of these six shapes.



How many of these shapes have two-thirds shaded?

- (A) 2
(B) 3
(C) 4
(D) 5

30. Jess started listening to her favourite songs. The minute hand on the clock was pointing to the 9.



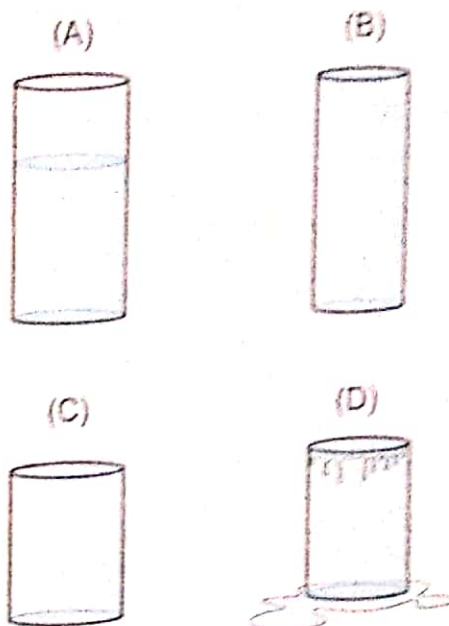
She finished listening less than an hour later. The minute hand was pointing to the 5.

For how long was Jess listening to her favourite songs?

- (A) 4 minutes
(B) 8 minutes
(C) 20 minutes
(D) 40 minutes

31. Sasi poured a full jug of water into each of these containers.

Which container can hold more than the jug?



32. There are two types of marbles. The small marbles are called 'taws' and the large marbles are called 'tommies'.



1 tommy is worth 3 taws.

Ravi has 6 tommies and 12 taws. He swaps half of his taws for tommies.

How many tommies does Ravi have after this?

- (A) 8
(B) 10
(C) 12
(D) 14

33. Tara is adding two numbers. Some of the digits are hidden by symbols.

$$\begin{array}{r} \star \text{ ?} \\ + 4 \star \\ \hline 103 \end{array}$$

The two digits hidden by stars are the same.

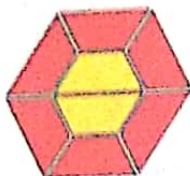
What digit is **?** hiding?

- (A) 1
- (B) 5
- (C) 7
- (D) 8

34. Tom had some blocks that were all the same size and shape. He used two of them to make this regular hexagon.



He placed six more blocks around this hexagon to make a bigger regular hexagon.



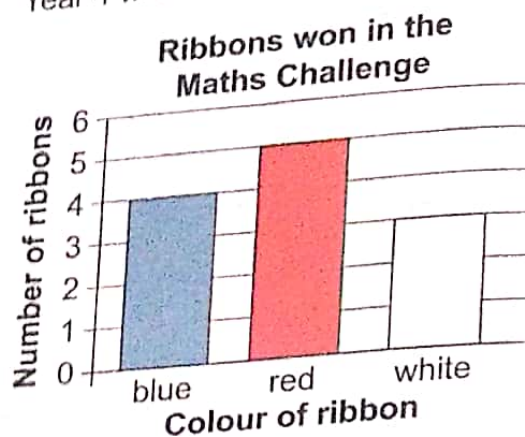
How many more blocks does he need to place around this shape to make the next bigger regular hexagon?

- (A) 6
- (B) 10
- (C) 12
- (D) 18

35. At Sam's school, points and ribbons are awarded for the first three places in each round of a Maths Challenge. The table shows the point value for each colour of ribbon.

Colour of ribbon	Number of points
blue	3
red	2
white	1

This graph shows the number of ribbons Year 4 won in the Maths Challenge.



What was the total number of points won by Year 4?

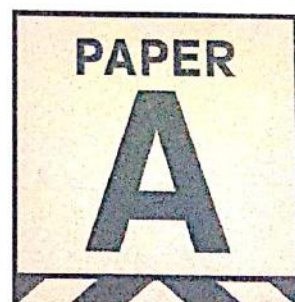
- (A) 5
- (B) 12
- (C) 23
- (D) 25

Yr 3 REACH Maths 2019.

- | | | | |
|-------|-------|-------|-------|
| 1. A | 11. C | 21. B | 31. A |
| 2. C | 12. A | 22. C | 32. A |
| 3. B | 13. D | 23. A | 33. D |
| 4. C | 14. A | 24. D | 34. B |
| 5. C | 15. A | 25. A | 35. D |
| 6. D | 16. B | 26. B | |
| 7. B | 17. C | 27. B | |
| 8. C | 18. D | 28. C | |
| 9. B | 19. A | 29. A | |
| 10. B | 20. B | 30. D | |

THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER

Australia ¹	Year 3
Brunei	Primary 3
Egypt	Year 3
Hong Kong	Primary 3
Indian Subcontinent ²	Class 3
Indonesia	Year 4
Malaysia	Standard 3
Middle East ³	Class 3
New Zealand/ Pacific ⁴	Year 4
Singapore	Primary 2
Southern Africa ⁵	Grade 3



¹ All international schools registered with UNSW Global (which have an 8-digit school code starting with 46) should sit the papers according to the Australian year levels.

² Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.

³ Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria, Jordan and Pakistan.

⁴ Pacific Region: Vanuatu, Papua New Guinea and Fiji.

⁵ Southern Africa Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia.



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